

Yi Zhen

Shanghai, China

🐙 <https://github.com/i-zhen>

in <https://linkedin.com/in/izhenyi>

🌐 <https://izhen.me>

+86 177 1768 7328

iamzhenyi@gmail.com

Summary

Senior Software Engineer and Tech Architect with extensive expertise in AI systems, LLM optimization, and high-performance compiler frameworks. Proven track record in leading technical teams to build large-scale, production-ready AI infrastructure, including LLM deployment, model compression, and operator development. Experienced in managing the full lifecycle of complex technical projects, from research-driven ideation to the implementation of open-source frameworks and university-enterprise collaborations. Expert in bridging the gap between cutting-edge AI research and robust software engineering, specializing in high-performance computing, smart contract compilers, and domain-specific languages.

Work Experience

- Ant Group** Shanghai, China
Senior Software Engineer, Team Lead & Tech Architect Mar. 2021 - Now
 - Architecting and building advanced AI systems focused on the optimization of LLM performance and operator development.
 - Leading a specialized engineering team to develop 'SIR', a pioneering open-source smart contract compiler framework on AntChainOpenLabs.
 - Managing the end-to-end development of the Gödel query language, a high-performance DSL for data processing.
 - Driving innovation through strategic university-enterprise cooperation projects with Sun Yat-sen University (SYSU).
- Huawei Research** Shanghai, China
Software Engineer, Tech Lead, Development Nov. 2018 - Mar. 2021
 - Served as Tech Lead for the design and implementation of a high-impact low-code development platform.
 - Acted as a core developer for the Open Ark compiler, contributing to advanced compilation technology.
- Citigroup** Shanghai, China
Software Engineer, Development, Continuous Integration Apr. 2017 - Nov. 2018
 - Engineered critical components for an anti-money laundering (AML) platform.
 - Developed and maintained the wire transfer system for block trade agents, ensuring high reliability and compliance.

Projects

- LLM Optimization of Training and Inference** Feb. 2024
Tech Lead
 - Driving LLM deployment and training strategies to maximize computational efficiency.

- Implementing model compression techniques and performing rigorous LLM performance profiling.
- Developing and tuning specialized operator libraries for high-performance LLM inference and evaluation.

• **Smart Intermediate Representation (SIR)**

Smart Contract Compiler Framework Co-founder *Nov. 2022 - Jan. 2024*

- Co-founded a universal technical solution designed to enhance safety and performance for smart contracts across various programming languages.
- Developed a framework to ensure multi-scenario compatibility for blockchain platforms.

• **Gödel Query Language**

Project Manager *Jul. 2021 - Oct. 2022*

- Developed a high-performance query language and compiler powered by a Datalog engine.
- Designed a Rust-like DSL for CodeFuse-Query, featuring strict type checking, type inference, and advanced error messaging.

• **ERC721G: Anti-Theft ERC721 Solution**

Co-founder *Jun. 2022 - Oct. 2022*

- Developed the world's first anti-theft NFT series solution, providing a universal technical layer to protect ERC721 smart contract assets.

• **PPrinter: A generic derivable Haskell pretty printer**

Author *Jun. 2016 - Aug. 2016*

- Created a Haskell library enabling automatic derivation of pretty-printing functions for arbitrary user-defined data types.

Sun Yat-sen University ASC Student Supercomputer Challenge

• **Team**

Team member *2014 - 2014*

- Mastered numerical methods, GPGPU heterogeneous computing, and multiprocessor programming.
- Optimized the SU 2 (Stanford University open-source C++ code for PDE analysis) and assisted in HPL performance benchmarks.

Education

• **University of Edinburgh**

Artificial Intelligence

Shanghai, China
Sep. 2015 - Nov. 2016

• **Sun Yat-sen University**

Software Engineering

Shanghai, China
Sep. 2011 - Jun. 2015

Technical Skills

LLM Optimization, LLM Deployment, Model Compression, Performance Profiling, AI Systems Architecture, Operator Development, Smart Contract Compiler Frameworks, Datalog, Rust, Python, C++, Haskell, C, CUDA, Triton, MLIR, LLVM, vLLM, SGLang, ONNXRuntime, Solidity, TypeScript, Scala, Java, Prolog, Coq, ASM, web3.js, ether.js, Hardhat, Tilelang, Project Management, Continuous Integration (CI/CD)